

ACC 2026 Tutorial on Bridging Control with Neural Network Verification

Part 2: Applications to Control

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What Applications can α, β -CROWN Solve?

- **Application 1: Bound Computation**
 - **Reachability analysis** [1,2]
- **Application 2: Non-linear Satisfiability Solver Over the Reals**
 - **Lyapunov analysis** [3,4]
 - **Contraction analysis** [5]
 - **Barrier function analysis** [6]
- **Application 3: Nonlinear Optimization**
 - **Model predictive control** [7]

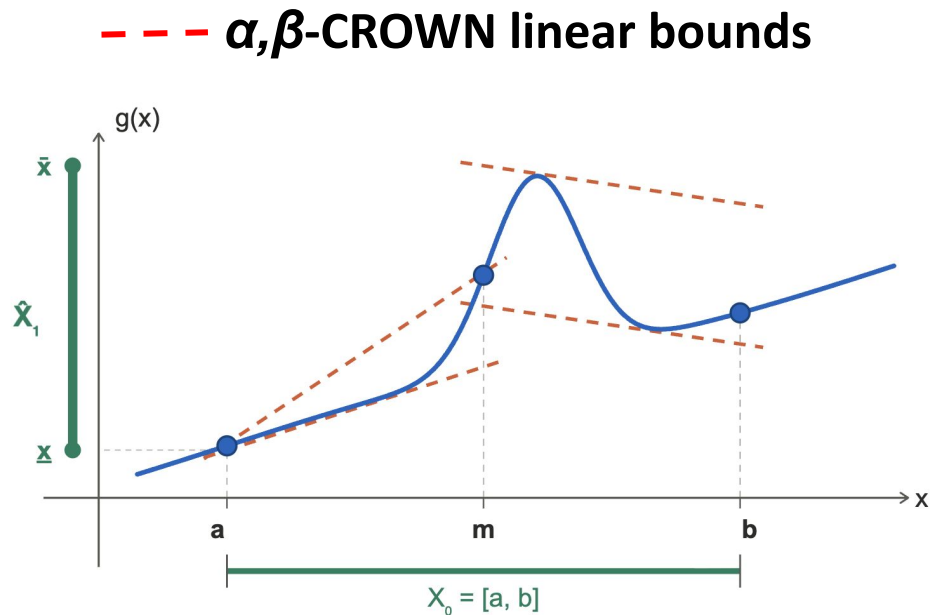
Application 1: Reachability Analysis

The question:

Given an initial box X_0 , and a discrete time system $x_{t+1} = g(x_t)$, for what states can the system reach?

What α, β -CROWN Provides:

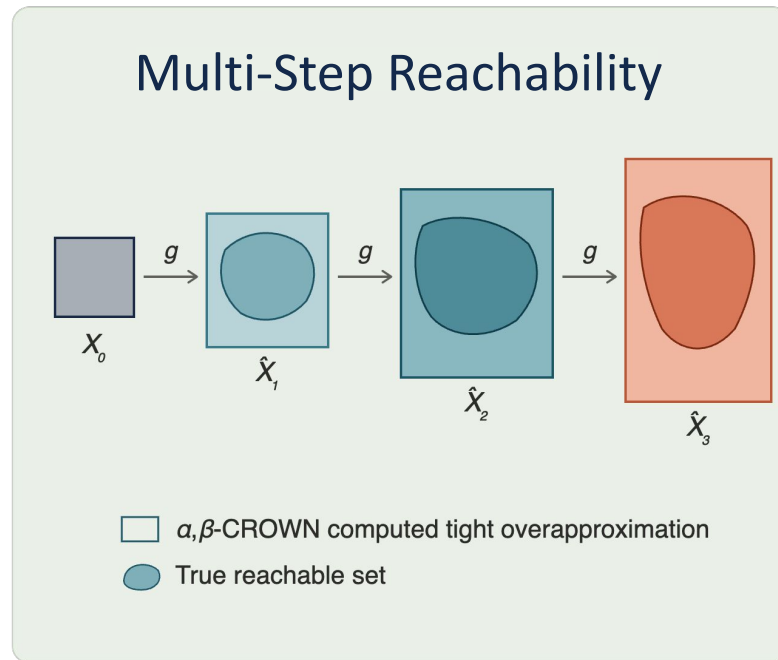
By treating g as a computation graph, the one step reachable set can be obtained from the **bounds returned by α, β -CROWN**.



Application 1: Reachability Analysis

Multi-Step:

- By composing the dynamics like $g(g(x))$, a multi-step reachability analysis for **nonlinear systems** can be done.
- On each Branch-and-Bound subdomain, a **tight linear bound will also be returned** in addition to the numeric bounds.



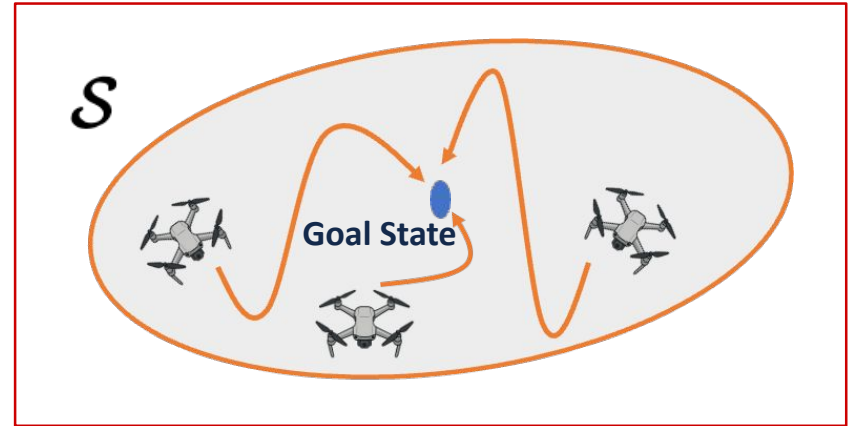
Application 2: Lyapunov Stability

The question:

For a given controlled system:

$$x_{t+1} = f(x_t, \pi(x_t)) = g(x_t)$$

Find and verify the region of attraction (ROA), a set $\mathcal{S}$ such that every trajectory starts within $\mathcal{S}$ converges to the equilibrium x_* .



Application 2: Lyapunov Stability

Lyapunov theorem:

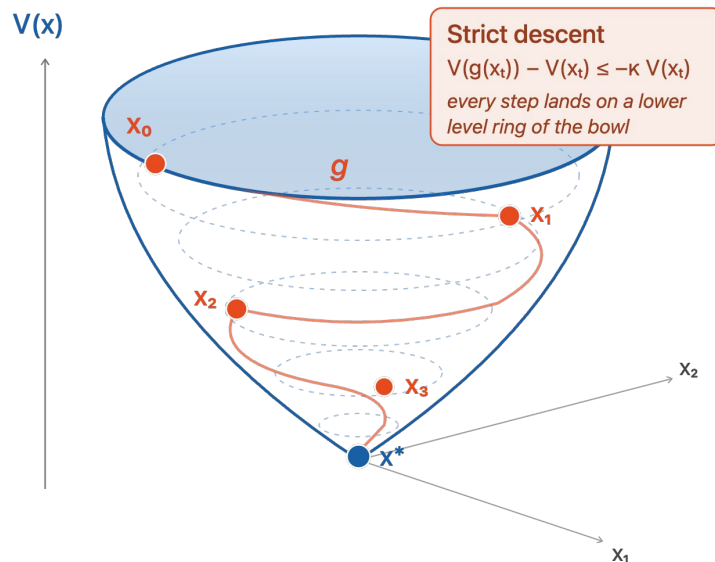
Given a forward invariant set $\mathcal{S}$, if there exists a function $V : \mathcal{S} \rightarrow \mathbb{R}$

$$V(x^*) = 0$$

$$V(x) > 0 \quad (\forall x \in \mathcal{S} \setminus \{x^*\})$$

$$V(g(x_t)) - V(x_t) \leq -\kappa V(x_t) \quad (\forall x_t \in \mathcal{S})$$

then $\mathcal{S}$ is a subset of the ROA. The first two conditions can be satisfied by construction [1,2]. We will focus on the third.

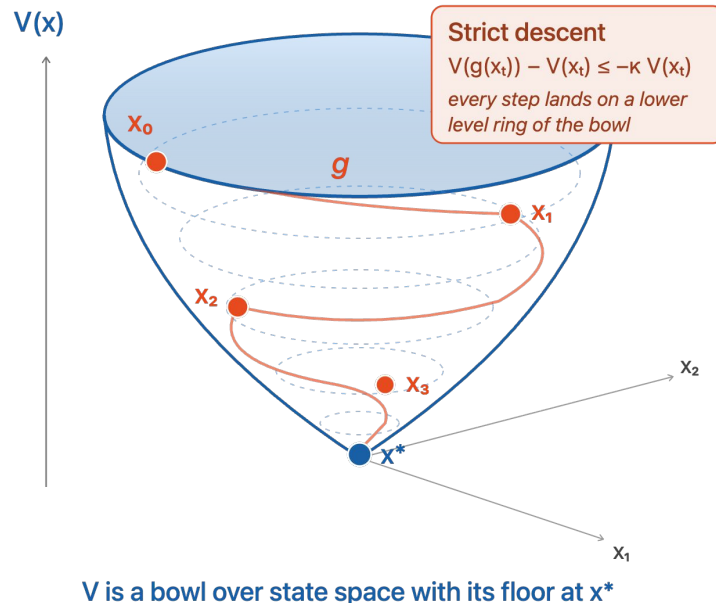


V is a bowl over state space with its floor at x^*

Application 2: Lyapunov Stability

Why α, β -CROWN:

- Lyapunov function, controller, and dynamics are all **highly nonlinear**.
- But the problem itself is still a **general satisfiability problem over nonlinear real arithmetic**.



Application 2: Key Challenge

A direct translation:

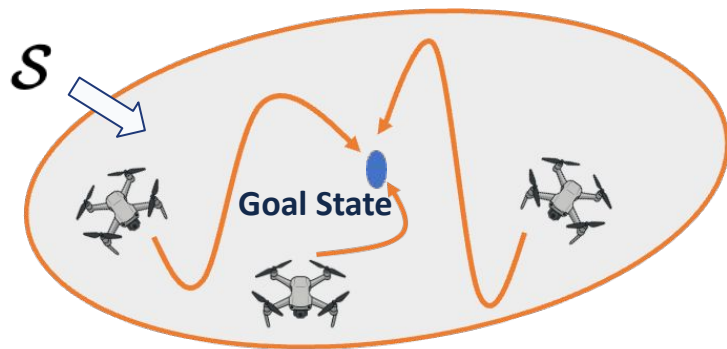
Denote $F(x) = V(g(x)) - (1 - \kappa)V(x)$,
and given this $\mathcal{S}$, a naive rewrite yield
the following satisfiability problem:

$$(-F(x) \geq 0) \wedge (g(x) \in \mathcal{S}), \quad \forall x \in \mathcal{S}$$



Not directly specifiable
in α, β -CROWN!

What control theory gives



- Verification checks **only**
needs to be done inside $\mathcal{S}$

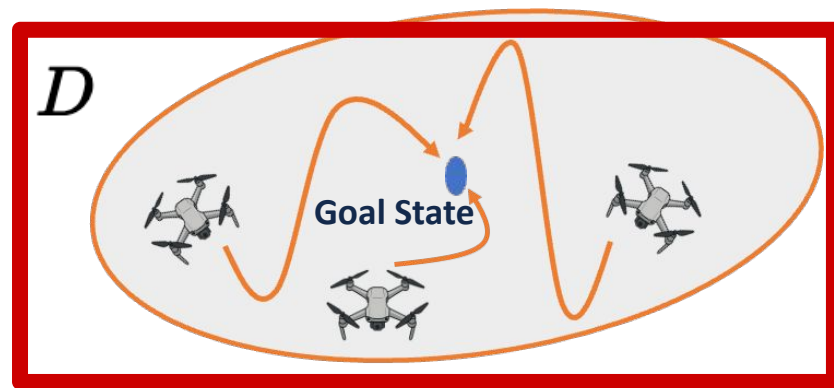
Application 2: Key Challenge

A naive fix:

Introduce the box $\mathcal{D}$, and directly convert the condition to be

$$-F(x) \geq 0, \forall x \in \mathcal{D}$$

And then find the maximal ρ sublevel-set so that $\{V(x) \leq \rho\} \subset \mathcal{D}$, then use this sublevel-set as the ROA.



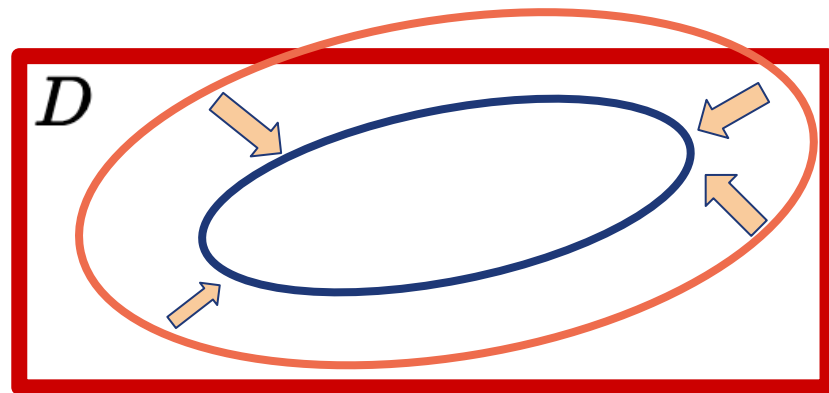
- Fully convert the verification into the box domain $\mathcal{D}$.

Application 2: Key Challenge

To ensure the verifiability of the conditions, we have to:

1. Pick a **conservative box** $\mathcal{D}$ so that the Lyapunov decrease condition holds over the full box.
2. Pick a **conservative sublevel set** to ensure $\{V(x) \leq \rho\} \subset \mathcal{D}$.

This significantly impacts the volume of the ROA obtained.



- Orange indicates true ROA, but we can only verify a **conservative estimation inside the blue curve**.

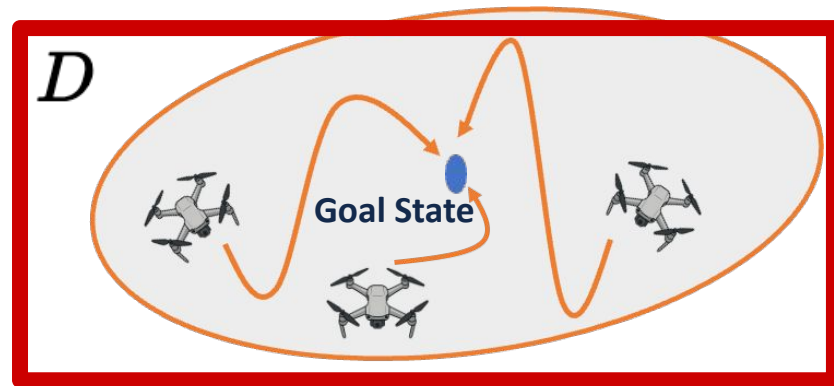
Application 2: Key Technique

The better solution:

We would rewrite to directly handle the intersection $\mathcal{D} \cap \{x : V(x) \leq \rho\}$.

Given this $\mathcal{D}$, we need to verify:

1. The **forward invariance** of the intersection $\mathcal{D} \cap \{x : V(x) \leq \rho\}$.
2. The **Lyapunov condition** holds on this same intersection.

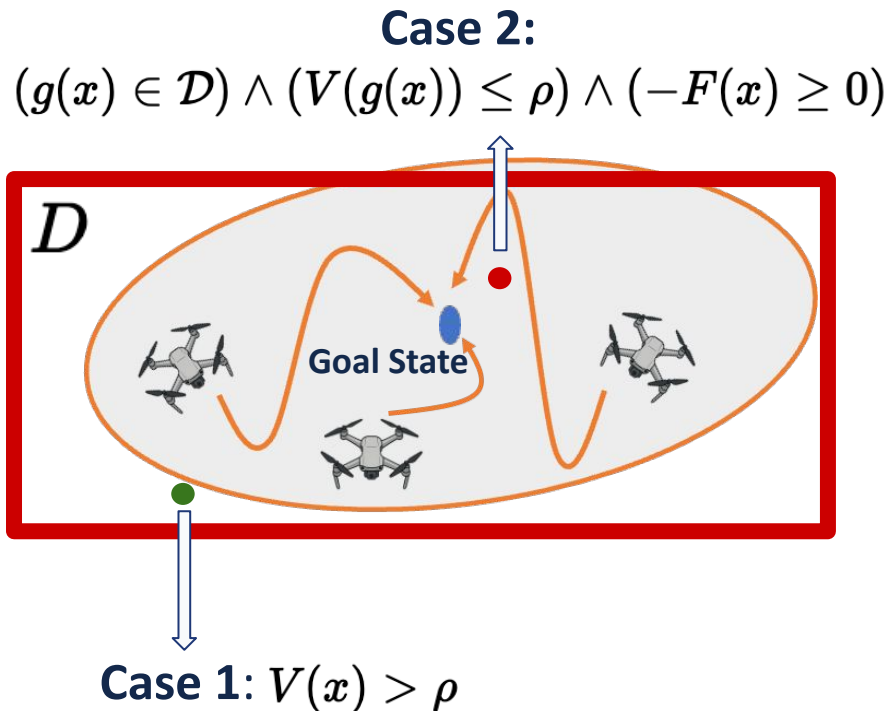


Application 2: Key Technique

The better solution:

For each point in domain $\mathcal{D}$, either:

1. It is **outside of the intersection** and we have $\{x : V(x) > \rho\}$.
2. It is **inside the intersection** $\mathcal{D} \cap \{x : V(x) \leq \rho\}$, Lyapunov condition holds, and the next state will remain in the intersection.



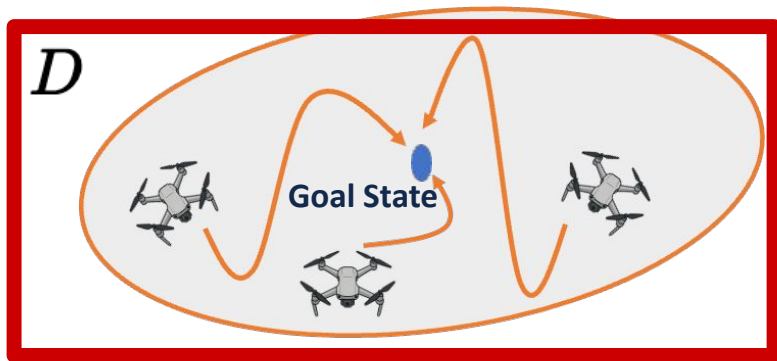
Application 2: Final Condition

Final Verification Condition:

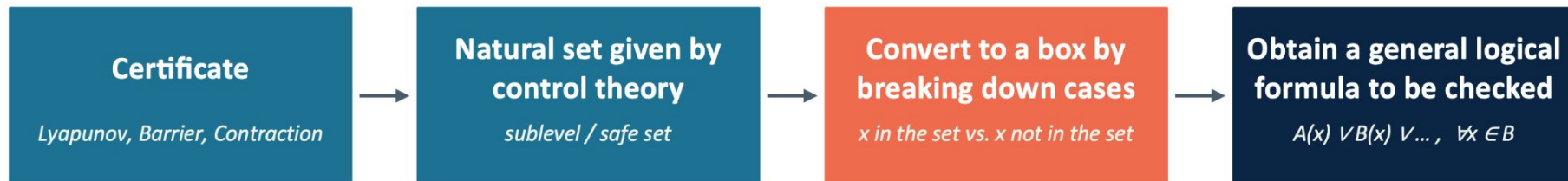
Denote $F(x) = V(g(x)) - (1 - \kappa)V(x)$,
if the condition

$$(-F(x) \geq 0 \wedge g(x) \in \mathcal{D}) \vee (V(x) > \rho), \quad \forall x \in \mathcal{D}$$

holds, then **the intersection** $\mathcal{D} \cap \{x : V(x) \leq \rho\}$
is a **subset of the ROA**. This satisfiability
problem can then be **checked efficiently by**
 α, β -CROWN.



Application 2: The Pattern



- The pipeline is **general**: We support **broad class of functions**, such as neural nets/sin/cos/tan/polynomials that are commonly used in control.
- The pipeline is **scalable**: The pipeline is **more scalable** than other general SMT solvers such as dReal. We also provide **compatible APIs** for dReal.
- The pipeline is **composable**: We can also **handle a diverse class of verification problems**, as long as it can be formulated into a satisfiability problem.

Application 3: MPC

The question (not computing bounds anymore):

Given a discrete-time dynamics $x_{t+1} = f(x_t, u_t)$, find an action sequence over horizon H that minimizes the total cost

$$\min_{u_0, u_{H-1}} \sum_{t=0}^{H-1} c(x_t, u_t)$$

What α, β -CROWN provides:

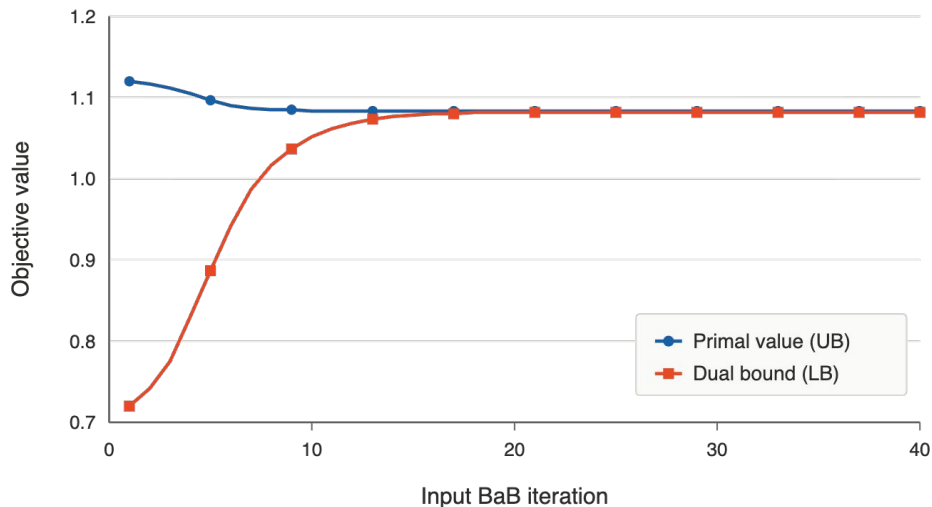
α, β -CROWN treats the cost function as a computation graph from $\mathcal{U}^H \rightarrow \mathbb{R}$ and will use the optimizer functionality to find an optimal solution for the problem.

Application 3: Optimality Guarantee

Benefits:

- Fully utilize the GPU parallel and acceleration as in BaB.
- Primal bound is **feasible MPC solution**, dual bound is α , β -CROWN lower bound, when they converge, we have **optimality guarantee**.

Optimization convergence from BaB history



Next: Code Examples of α, β -CROWN
<https://abcrown.org/tutorial>